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A multi-objective optimization of ballistic and cruise missile fire plans based on damage calculations from missile impacts on an airfield defended by an air defense artillery network

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Abstract

This paper presents a new methodology for optimizing missile fire plan parameters in order to maximize the damage caused to an airfield by the impacts of ballistic and cruise missiles that leak through an air defense artillery network using weaponeering calculations. This new methodology models the damage to the airfield runway, fuel infrastructure, and parked aircraft caused by both unitary and submunition warheads carried by the ballistic and cruise missiles. It then uses the airfield damage models paired with an air defense simulation in a multi-objective optimization to find the non-dominated frontier of missile fire plan parameters that inflict the greatest damage to the different parts of the defended airfield. It demonstrates the methodology for three increasingly sized ballistic and cruise missile fire plans.

Keywords

Air and missile defense, multi-objective optimization, weaponeering, airfield damage

1. Introduction

Almost every modern military maintains an arsenal of theater ballistic missiles (TBMs) and cruise missiles (CMs). These missiles provide a long-range strike capability in order to shape the battlefield for future engagements. Due to their historically high expense and complex nature, TBMs and CMs have typically been used in small numbers and always in uniform raids made up of only a few TBMs of the same type or only a few CMs of the same type.[1] This trend will not continue.

The proliferation of TBM and CM technology is accelerating, leading to growing stockpiles of these missiles.[2,3] Soon, many countries will be able to attack with large raids made up of many different types of missiles landing simultaneously onto a target area. These large and mixed raids open the opportunity for the attacking missile force to optimize missile raids, since they can tailor the types, quantities, and attack parameters of the missiles in the raid to achieve certain effects.

The fire plan specifies the number, type, and attack parameters of each TBM or CM to shoot in a single raid against a specific target. In Air Defense Artillery (ADA) analysis efforts, this fire plan must be considered before analyzing the ADA systems, since the ADA systems are defensive in nature. However, in the majority of the current ADA analysis efforts,[4–8] this fire plan is either not initially optimized or designed to maximize the number of leakers – the incoming TBM and CM threats that are not shot down by the ADA platforms and impact the target. In the past, maximizing the number of leakers was sufficient because with small and uniform raids, the amount of

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damage to the target was highly correlated to the number of leakers. In the future, this may not be the case due to the differences in accuracy, lethality, and vulnerability between TBMs and CMs and between different types of TBMs and CMs.[9,10] This diversity means that the capability of the ADA systems to defend against each missile type is different, and the effects of leakers from each missile type are also different. A new methodology is needed to develop and optimize missile fire plans with a mix of different types of both TBMs and CMs based on the damage caused by the leakers and not simply the total number of leakers.

Brown et al.[11] moved toward this with their two-sided optimization strategy that, from the attacker's perspective, maximized the destruction caused by missile impacts while simultaneously, from the defender's perspective, minimized the destruction. Their analysis only considered ballistic missiles, and used a subjective value score for each target based on its criticality, vulnerability, and reconstitutability. While not a true damage effects model based on weaponeering, it should be commended for being one of the first ADA analysis efforts to optimize the enemy fire plan before considering the defense.

This work will improve upon the current methods by developing an effects-based approach for creating missile fire plans, applicable to both TBMs and CMs, that maximizes the damage to various parts of an airfield. This methodology will then be paired with an ADA simulation and a multi-objective optimization to find the optimal set of missile fire plan parameters that maximize the damage to the different parts of the airfield.

2. Damage effects model methodology

One of the primary targets for TBMs and CMs are airfields. The targets on an airbase can be grouped into runway or airfield targets, as shown in Figure 1. These targets are static, making them more easily targeted by long-range missiles. For an attack on the runway, the missile force would like to damage the runway to such an extent that the number of takeoffs and landings is reduced or eliminated for a period of time. An attack on the airfield will target the fuel sites or the parked aircraft. In an attack on the fuel sites, commonly called Petroleum, Oil, and Lubrication (POL) points, the attacking missile force wants to also reduce or eliminate the number of takeoffs by interrupting aircraft operations due to a lack of fuel or a bottleneck of aircraft trying to use the remaining undamaged POL points. Finally, the attacking missile force may choose to directly target the parked aircraft in order to destroy them before they can be used in combat. While this is more advantageous than temporarily slowing or halting operations, it is more difficult because the aircraft

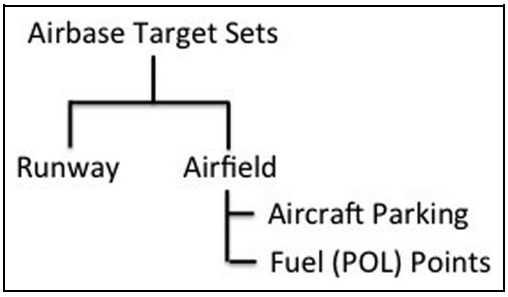


Figure 1. Airbase target architecture. POL: Petroleum, Oil, and Lubrication.

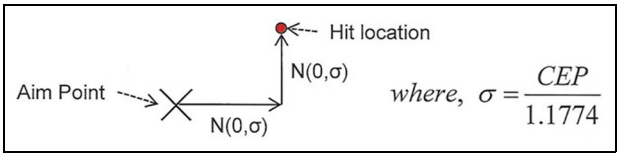


Figure 2. Impact point calculation method.

are continually repositioning on the airfield due to missions or maintenance. The following sections will develop a damage model for each of these areas of the airbase.

2.1 Missile impact error

The accuracy of a TBM or CM is quantified by its circular error probable (CEP). The CEP is the radius of a circle, centered at the aim point, in which 50% of the weapons are expected to land.[12] The impact error can also be expressed as range and deflection errors. If the missile is not biased to miss in any given direction, then the mean impact point will be the aim point, and the range or deflection error can be described as $N(0,\sigma_r)$ or $N(0,\sigma_d)$. If the weapon is also likely to miss an equal amount in range and deflection (the standard deviations are equal and uncorrelated), then the impact distribution can be described by the Rayleigh distribution. A careful manipulation of the Rayleigh distribution equation reduces to the following[12]:

$$CEP = 1.1774\sigma$$

This form of the CEP is easier to work with in simulation. It can be used to calculate a miss distance in both range and deflection in order to calculate the impact point, as shown in Figure 2.

For an unguided missile, this method is used directly to calculate the impact point for both TBMs and CMs. For guided missiles, the terminal guidance phase will fall into one of three categories: direct hit, near miss, or gross error.[12] Most weapons' near miss patterns exhibit a normal

Table 1. Notional missile guidance parameters.

Missile	Precision/non-precision	CEP (m)	Probability of direct hit	Probability of failure
TBM1/TBM2	Non-precision	200	N/A	N/A
TBM3	Non-precision	50	N/A	N/A
CM1	Precision	20	50%	5%
CM2	Precision	20	70%	3%

CEP: circular error probable; TBM: theater ballistic missile; CM: cruise missile.

distribution, $N(0,\sigma)$.[12] A gross error occurs when the guidance system fails and the weapon lands very far from its intended aim point. Typically, any impact more than four standard deviations from the aim point is considered a gross error.[12] Each of these outcomes will have a probability associated with it that is based on testing or simulation. In order to determine which of these three states the guidance system is in when it impacts, a random draw is conducted relative to the uniform distribution, $U(0,1)$, against the associated guidance probabilities for that missile.[12]

2.2 Warhead types

TBMs and CMs typically attack a target with two different types of payloads: unitary warheads or submunitions. A unitary warhead has a single explosive payload, typically tied to either a point or a proximity fuse. These warheads deliver all of the explosive power to a single location on the ground. A submunition warhead, sometimes referred to as a cluster munition, dispenses many smaller ''bomblets'' that disperse as they fall to the ground. These submunitions each have a small warhead tied to a proximity, point, or delay fuse. This type of warhead delivers the explosive power dispersed over a large area.

In an attack on a runway, a specific type of submunition is often used called a runway-penetrating munition. In order to create the largest possible crater, the detonation of the explosive should be subsurface.[13,14] Therefore, these munitions are designed to maintain a higher velocity as they fall and are shaped to burrow into the concrete on impact. A delay fuse is used to detonate the explosive once the penetrator has burrowed to a certain depth, defined by a time after impact with the surface. Because the detonation is subsurface, there is generally only a small amount of damage to surface targets away from the blast site because most of the overpressure and fragmentation is contained by the earth or pavement.

The release altitude of the submunitions affects the spread of the submunitions around the mean impact point. The release altitude also serves as a defense measure against the ADA systems. Generally, the submunitions are too small and numerous to be engaged by the ADA systems. If they are released before the ADA system can engage the missile carrier, then the submunitions propagate to the ground unimpeded. This must be balanced with the desired concentration of the submunitions. If they are released too high above the target, the resulting spread of the munitions can dilute the effectiveness of the submunitions.

The example case in this work will use three TBMs and two CMs as an illustrative example. These missiles are notional and do not represent any known system. TBM1 and TBM2 are the same missile with two different warheads: TBM1 is unitary and TBM2 contains 70 runway-penetrating munitions. TBM3 is a more accurate unitary missile. For the CMs, CM1 is a less accurate CM with 40 submunitions and CM2 is a more accurate CM with a unitary warhead. The guidance and accuracy parameters of all five missiles are shown in Table 1. These values reflect the general trend of CMs being more accurate than TBMs.[15]

2.3 Runway damage model

The overall damage to a runway caused by a series of TBM and CM attacks with different warheads can be calculated with a runway damage model. This overall model is composed of five main submodels: the missile flight propagation submodel; the penetrator flight propagation submodel; the concrete penetration submodel; the cratering submodel; and the runway availability submodel. These submodels apply to both TBMs and CMs for penetrator submunition warheads; however, the unitary warhead missiles do not use the penetrator flight or the concrete penetration models since they do not release penetrators. A diagram illustrating the overall methodology to include the submodels is illustrated in Figure 3.

2.3.1 Missile flight submodel. The missile flight propagation model is the first submodel in the runway damage model. This is a three-degree-of-freedom (3-DOF) model that propagates the TBMs and CMs from their launch point to their impact point for unitary munitions, or to their release point for missiles with submunitions. Its primary output is the orientation of the missile at impact for unitary munitions

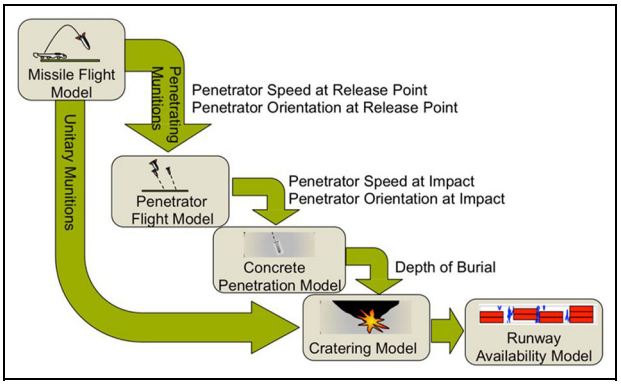


Figure 3. Runway damage model with submodels.

or the orientation and speed of the missile carrier at the release point for missiles with submunitions.

For TBMs, a 3-DOF model is used because the TBM reentry vehicles (RVs) are typically axisymmetric about the roll axis and aligned to the velocity vector with little to no yaw perturbations during their ballistic flight after the motor cutoff. This means that the roll axis is inconsequential and the pitch and yaw can be determined from the angle of the velocity vector. A simple 3-DOF model is derived by Forden[16] and is shown below in vector form:

$$\bar{F} = m(t)\bar{g} - 2m(t)(\bar{w}_{\oplus} \times \bar{v}) + m(t)\bar{w}_{\oplus} \times \bar{r} \times \bar{w}_{\oplus} + \bar{T} - 0.5\rho(t)C_d A v\bar{v}$$

In this model, all vectors are with respect to the inertial frame. The term $m(t)$ is the mass as a function of time due to fuel burn, $\bar{w}_{\oplus}$ is the earth's rotational velocity vector, $\bar{v}$ is the rocket velocity vector, $\bar{r}$ is the position vector, $\bar{T}$ is the thrust vector, $\rho(t)$ is the density as a function of time, C_d is the coefficient of drag, A is the body cross-sectional area, and v is the scalar velocity.

The speed of the TBM RV at either the release altitude or the impact point is found by determining the magnitude of the velocity vector. The orientation of the TBM RV at the release altitude or the impact point is found by computing the dot product of the velocity vector and the ground normal vector and dividing by the scalar velocity. These values combined with any penetrator ejection angle form the initial conditions for the penetrator flight model. For CMs, a standard aerodynamic trim program[17] can be used to calculate the pitch, roll, and yaw for steady flight at a given airspeed. These angles and speeds will be the initial conditions at the release point or the conditions at the impact point.

2.3.2 Penetrator flight submodel. The penetrator flight submodel requires additional degrees of freedom beyond that of a 3-DOF model. The penetrators are typically ejected at a small angle away from the missile carrier longitudinal axis in order to facilitate their larger coverage over the ground. This means that, initially, the penetrator longitudinal axis will not align to that of the velocity vector. This creates a yaw of repose[18] between the penetrator's longitudinal axis and its velocity vector. This off-axis flight cannot be accounted for with a 3-DOF model. In addition, the cratering submodel is sensitive to the penetrator's depth, which is sensitive to the impact velocity and angle, which is affected by the yaw of repose. Therefore, additional accuracy is required beyond what is provided by a 3-DOF model.

These factors can be accounted for with either a six-degree-of-freedom (6-DOF) model or a modified point-mass model (MPM).[18,19] In general, the 6-DOF model will be more accurate than the MPM, but comes at the cost of added computational resources. The MPM has been shown to nearly match the 6-DOF solutions except in cases where a ballistic projectile is fired at an extremely high quadrant angle.[19] This is because the MPM neglects the epicycle yawing and pitching motion and assumes that the yaw of repose changes slowly. At apogee for a projectile fired at a large quadrant angle, the yaw of repose changes rapidly and the epicycle motions can become significant. This is not the case for a penetrator munition that is ejected from a missile well beyond its apogee. Therefore, this work made use of the MPM. The vector equations for the MPM for the ballistic flight for a non-spinning, fin-stabilized projectile are as follows[19]:

$$\frac{d\vec{V}}{dt} = \frac{\rho S C_D}{2m} v\vec{v} + \frac{\rho S C_{L_\alpha}}{2m} v^2 \vec{\alpha}_R + \vec{g}$$

$$\vec{\alpha}_R = \frac{(C_{M_q} d)}{(C_{M_\alpha} v^4)} \left(\vec{v} \times \left(\vec{v} \times \vec{g}\right)\right)$$

$$\vec{v} = \vec{V} - \vec{W}$$

$$\alpha_t = \sin^{-1}\left|\vec{\alpha}_R\right|$$

where V is the velocity vector of the penetrator with respect to the inertial frame, W is the wind vector with respect to the inertial frame, ρ is the air density, g is the gravity vector, and d, S, and m are the projectile reference diameter, reference area, and mass, respectively. For the aerodynamic coefficients, C_D and $C_{L\alpha}$ are the coefficients of drag and lift, C_{Ma} is the pitching moment coefficient, and C_{Mq} is the pitching dampening moment coefficient. Finally, α_t is the total angle attack between the velocity vector and body x-axis that is used in conjunction with the Mach number when determining the aerodynamic coefficients. In these equations, the Coriolis acceleration is ignored due to the small lateral distances that the penetrators will typically travel. The example cases of this paper

modeled the submunitions using the aerodynamic data of a 120 mm mortar.[19]

These MPM equations are solved in a time-step fashion from penetrator ejection to impact. The initial conditions are determined based on the orientation and speed of the missile carrier at the submunition's release altitude plus any additional ejection angle imparted from the submunition carrier mechanism. The impact speed is calculated as the magnitude of the velocity vector at impact, and the angle between the concrete surface's normal vector, and the penetrator's velocity vector is combined with the yaw of repose to determine the impact angle. These form the initial conditions for the concrete penetration submodel.

2.3.3 Concrete penetration submodel. The concrete penetration submodel determines the depth that the submunition burrows into the concrete of the runway, given its impact velocity and orientation. Concrete has a complicated nonlinear response to impact loading.[20] As a result, there is not a closed form solution for the penetration depth. The penetration depth can be determined with either finite element analysis or empirical formulas. Although finite element analysis is more accurate than empirical formulas, the results of the empirical formulas can be calculated much faster and can capture a wide range of impact and target conditions that would be difficult for a single numerical model.[21] For these reasons, the empirical models were used in this work. A comparison of three empirical models to experimental data[22] found that the Sandia empirical equations predicted the test data most accurately. The Sandia concrete penetration equations are given below in SI units[12,23]:

$$x = 0.0008\, SNk_h \left(\frac{m}{A}\right)^{0.7} \ln(1 + 0.000215V^2) \text{ for } V < 61\frac{m}{s}$$

$$x = 0.000018\, SNk_h \left(\frac{m}{A}\right)^{0.7} (V - 30.5) \text{ for } V \geq 61\frac{m}{s}$$

In these equations, x is the penetration depth, m is the penetrator mass, A is the average penetrator cross-sectional area, V is the impact velocity, S is a dimensionless penetrability factor (explained below), N is the nose factor (explained below), and K_h is the lightweight penetrator factor that can be calculated as follows[23]:

$$k_h = 0.46m^{0.15} \text{ for } m \leq 182\text{kg}$$

$$k_h = 1.0 \text{ for } m > 182\text{kg}$$

The nose factor, N, in the penetration depth equations can be found by measuring the penetrator's Caliber-Radius-Head (CRH). The CRH is a dimensionless number that describes a shell's ballistic length to its radius of curvature for its nose. The CRH number for most munitions is formally shown as a dual number, such as 3/4CRH, where the first number is the ballistic length and the second number is the radius of curvature; however, many times the ballistic length and the radius of curvature are equal. When this is the case, the CRH is commonly abbreviated to just a single number such as 4CRH.[24] The nose factor, N, can be found by the equations below[23,25]:

$$N = 0.18 * (CRH - 0.25)^{0.5} + 0.56$$

Finally, the target material S-number in penetration depth equations is defined as shown below, where P is the volumetric percent of rebar (expressed as a percentage, that is, 2% = 2.0), T_c is the target thickness in calibers (if $T_c > 6$, then set $T_c = 6$), t_c is the concrete cure time as a fraction of a year (if $t_c > 1$, then set $t_c = 1$), f_c is the concrete compressive strength, K_e is the concrete edge factor, W_1 is the target width in calibers of the penetrator, and F is 20 for reinforced concrete or 30 for unreinforced concrete[23]:

$$S = 0.085K_e(11 - P)(t_c T_c)^{-0.06} \left(\frac{35}{f_c}\right)^{0.3}$$

$$K_e = \left(\frac{F}{W_1}\right)^{0.3} \text{ for } W_1 < F$$

$$K_e = 1.0 \text{ for } W_1 \geq F$$

In the example case of this work, each submunition included a delay fuse with a depth range as given in Table 2. A range of depths was used since the time-delay fuse has inherent tolerances that equate to fluctuations in the depth at which it detonates. If a submunition's slant-line depth exceeded the distance determined by a random draw from within this range, then the submunition detonated at the slant-line depth specified by the random draw. However, if the submunition did not penetrate to this slant-line depth, then it detonated at the slant-line depth specified by the penetration depth.

2.3.4 Explosive cratering submodel. Both unitary and submunition warheads will cause cratering on the runway. When measuring the size of the explosively formed craters, there are normally two measurements given for every dimension: the apparent and true dimension. This is because material from the medium in which the explosion took place is initially ejected upwards and then some will fall back into the crater.[26]

The two critical factors in crater formation are the explosive mass and the depth of burst (DOB). For a given amount of explosive, the diameter of the crater will

Table 2. Notional unitary warhead parameters.

Missile	Warhead type	Number of penetrators	Delay fuse depth (inches)	Explosive weight (lbs)	Case mass to charge mass ratio
TBM1	Unitary	N/A	N/A	365	5
TBM2	Submunition	70	24–28	3	5
TBM3	Unitary	N/A	N/A	100	5
CM1	Submunition	40	18–22	1.5	5
CM2	Unitary	N/A	N/A	60	5

TBM: theater ballistic missile; CM: cruise missile.

increase with greater DOB up to a certain point. Beyond that point, the diameter of the crater decreases due to a larger portion of the energy from the explosion being absorbed by the earth. Likewise, the true depth of the crater will increase with DOB until a certain point where most of the energy is absorbed by the medium. Beyond that depth, the explosive will form a buried cavern called a camouflet. When a camouflet is formed, there is a small entry hole with a void below the surface.[27]

In a submunition warhead, the explosive charge is encased in a protective shell. Because this shell must be strong enough to maintain its shape during impact and tunneling through concrete, it will absorb a significant portion of the energy from the explosives. The amount of energy that is absorbed is described by the Modified Fano Equation[12]:

$$W_e = W(EEW)\left(0.6 + \frac{0.4}{1+2\frac{M}{C}}\right)$$

where W is the explosive weight, W_e is the equivalent weight of uncased explosive available after the case fracture, EEW is the equivalent explosive weight factor, M is the case mass, and C is the charge mass. From this equation, it is apparent that the fraction of available energy quickly decays as the case mass is increased and approaches a limiting fraction of 60% of the energy available for large case mass to explosive ratios. The EEW factor scales different explosives to an equivalent weight of TNT. The EEW factors for several common explosives can be found in Driels[12] or Maienschein.[28]

Research[12,13] into cratering has found that the crater dimensions scale with the cube-root of the equivalent uncased explosive weight, W_e. This is known as Hopkinson scaling. Thus, the crater radius and depth are given by the following:

$$r_c = W_e^{\frac{1}{3}}(r) \text{ and } d_c = W_e^{\frac{1}{3}}(d)$$

where r and d are the scaled radius and depth of the crater, r_c and d_c are the actual radius and depth of the crater, and W_e is the equivalent weight of explosive available after case fracture. These dimensions are based on the scaled DOB that is calculated with the equivalent uncased explosive weight, W_e, and the actual DOB, DOB_c[12]:

$$DOB = \frac{DOB_c}{W_e^{1/3}}$$

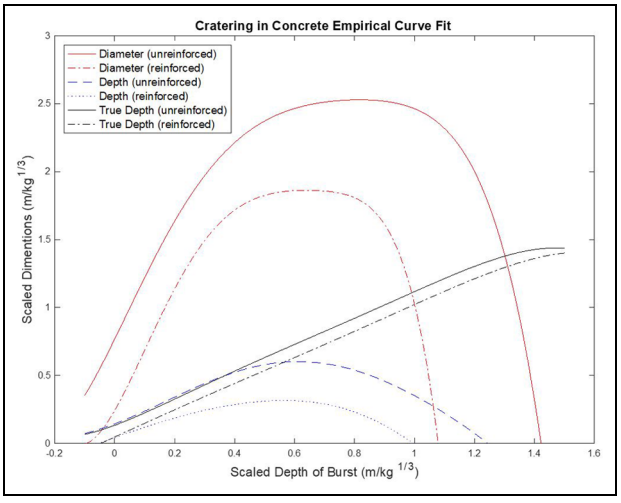


Figure 4. Crater scaled diameter and depth as a function of scaled explosive depth of burst (created based on images in Kiger and Balsara[14] and Driels[12]).

For a penetrating munition, the actual DOB (DOB_c) is the vertical depth of the submunition after its slant travel through the concrete. It is found using trigonometry from the concrete impact angle and the slant travel distance of the concrete penetration submodel. Using these scaling terms, empirical curve-fits of test data have been developed that predict the size of a crater within concrete. These are plotted in Figure 4 based on images in Kiger and Balsara[14] and Driels.[12] For each set of curves, the top is for unreinforced concrete and the bottom is for reinforced concrete. For surface burst munitions, the penetration depth is zero and the crater size is calculated by regressing the scaled diameter lines back to a depth of zero. For the five missiles used in the example case, the

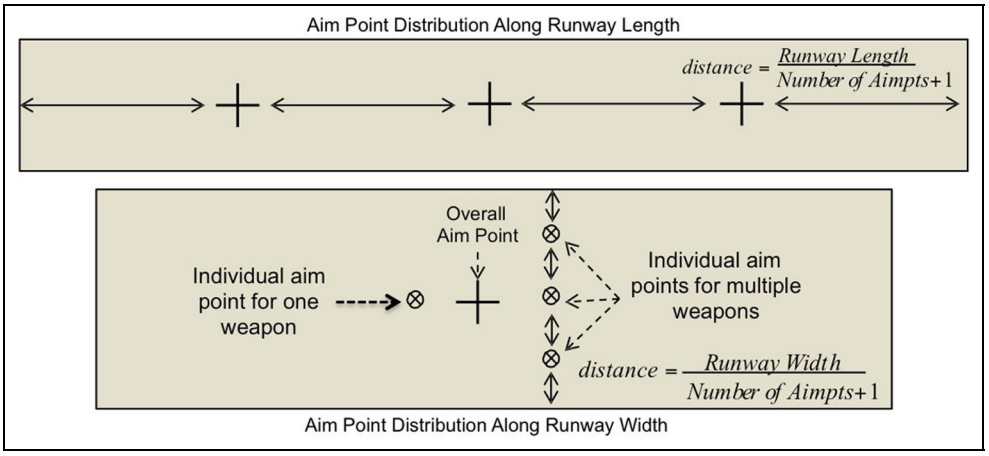


Figure 5. Aim point distribution along the runway and across the runway.

warhead types and sizes are explained in Table 2. These values were fabricated by the author and do not represent any known systems.

2.3.5 Runway damage submodel. Aircraft typically do not need the entire length and width of a runway to takeoff and land. The smallest strip of runway that an aircraft needs to either takeoff or land is called a minimum operating strip (MOS). A MOS is rectangular in shape and its dimensions are different for each aircraft type, operating conditions, and loading.[29] The MOS must be completely clear or all craters and debris for its full length and width and the MOS must completely lie within the bounds of the runway. However, it does not have to start or end at the beginning or ending of the runway.

When a missile force is targeting the runway with a set of accurate weapons, they will attempt to disperse those weapons along the length of the runway in order to close as many MOSs as possible. An example of the aim point distribution along the runway is shown in Figure 5. In this figure, the top diagram shows a methodology to achieve an even distribution of aim points along a runway. The bottom diagram shows the distribution of aim points along the width of the runway for either a single missile or for multiple missiles aimed at a single point along the length of the runway.

If the airbase is defended by ADA assets, those assets will not know the individual targets for each missile. Therefore, if a set of missiles is inbound to a set of aim points, but some of those missiles are shot down, then the remaining leakers will impact a random subset of the original aim points. For guided or unguided unitary TBMs or CMs, the impact point for each missile relative to its individual aim point is determined based on the weapon's CEP and guidance probabilities. For submunition warheads, the mean impact point of all the submunitions is determined in the same manner. Then, the individual impact points for each submunition can be determined around that mean impact point using the penetrator flight submodule.

2.3.6 MOS availability submodel. The final submodel of the runway damage model is used to determine the maximum number of MOSs of a given size that can occupy the undamaged portions of the runway. The most logical search pattern is to start at one end of the runway and work along the length to the opposite end. At the start location, place a potential MOS rectangle of the given size at the lower left corner of the runway. Then look inside the proposed MOS to determine if there are any craters or preexisting MOSs that lie within the bounds of the proposed MOS. If there are no craters or portions of another MOS, then that MOS location is good. Record the location and move to the top of that MOS and repeat the process. If there are either craters or MOSs within the proposed MOS location, then move to either the top of the crater furthest from the bottom of the proposed MOS location or the top of the preexisting MOS, whichever is further away from the bottom of the proposed MOS location. Repeat this process until the top of the proposed MOS is at above the top edge of the runway.

Once at the top edge of the runway, the next vertical search must start at some horizontal distance along the length of the runway. To find this distance, look at all craters and preexisting MOSs found during the last vertical search. Then, find the right-hand edge of these craters and preexisting MOSs. Move to the right-hand edge of the crater or preexisting MOS that is closest to the current proposed MOS start location and repeat the vertical search process. This process of searching vertically and then

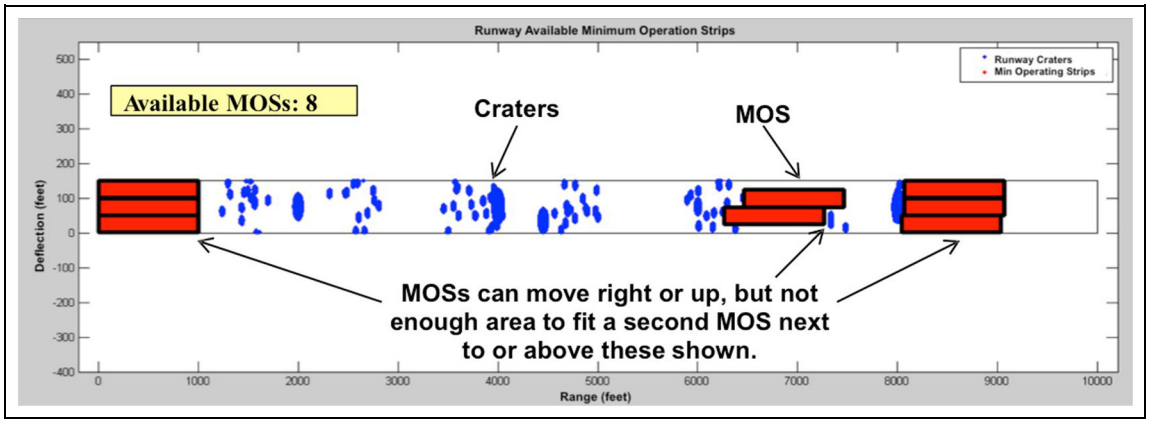


Figure 6. Sample results from the minimum operating strip (MOS) selection program.

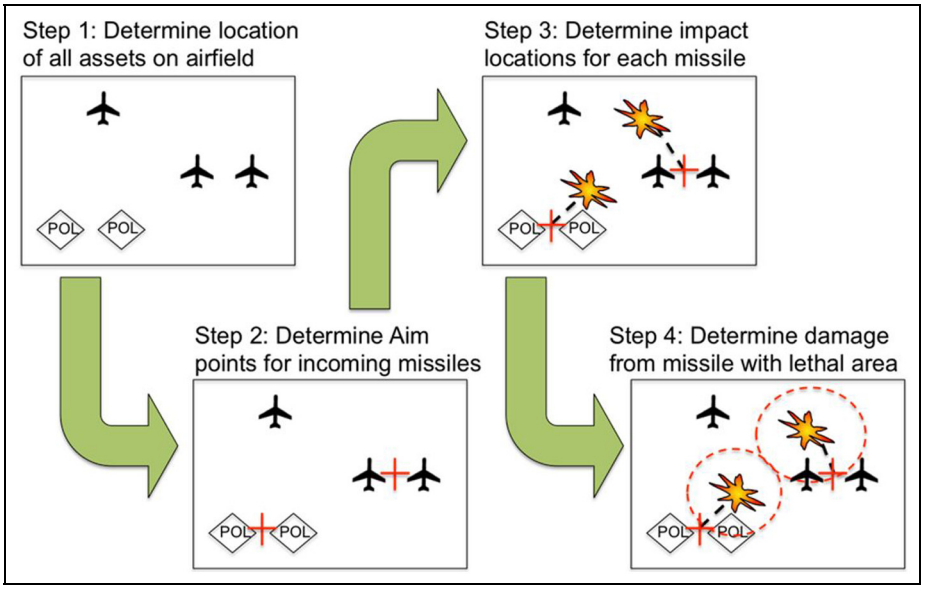


Figure 7. Airfield damage model process. POL: Petroleum, Oil, and Lubrication.

horizontally continues until reaching the far end of the runway. Once complete, the MOSs are counted.

This method produces a set of MOSs that is anchored at the lower left-hand corner of an available space. The MOSs could move up or right until they encounter another MOS or crater, as illustrated in Figure 6. However, there is not enough room either above or to the right of the MOSs to place another entire MOS of the given size.

3. Airfield infrastructure and aircraft damage model

Referring back to Figure 1, the second set of targets on an airbase lie off the runway and inside the rest of the airfield. These targets are the POL points and the aircraft parking locations. The methodology to determine the damage done to the airfield POL points and parked aircraft is summarized in Figure 7. The POL points are fixed locations that are known at the start of the attack. The aircraft, however, can reposition on the airfield to new parking locations. Therefore, the first step in the simulation is to determine the location of the aircraft parked on the ramp relative to the parking locations.

An airfield commander will want to maximize the distance between parked aircraft so that a single missile has a reduced chance of damaging multiple aircraft; however, aircraft of the same type are generally grouped together in assigned parking locations to facilitate maintenance and inspections. Thus, when determining the location of the aircraft on the airfield, the analyst should randomly place

Table 3. Notional lethal radius for missile–target pairs.

Missile	Lethal radius (m)			
	AC1	AC2	AC3	POL
TBM1	80	100	30	30
TBM3	60	80	20	20
CM2	40	50	20	15

TBM: theater ballistic missile; CM: cruise missile; AC: aircraft; POL: Petroleum, Oil, and Lubrication.

Table 4. Notional blue interceptor characteristics.

System	Missile	Interceptors per launcher	Max # in flight	Average velocity (m/s)	Seeker	Target set
Lower	Interceptor A	12	12	1400	Active	TBM/CM
	Interceptor B	8	10	1000	Semi-active	TBM/CM
	Interceptor C	4	5	1200	Semi-active	TBM/CM
Upper	UT Interceptor	8	5	2000	Active	TBM

UT: Upper Tier; TBM: theater ballistic missile; CM: cruise missile.

the first aircraft into a parking location for its assigned type. Subsequent aircraft should then be placed in parking locations for their type such that the minimum distance between all aircraft is maximized. If a certain aircraft type has more aircraft present than assigned parking locations, then it should be assigned to any parking location such that the minimum distance between all aircraft is maximized.

The second step is to determine the aim points relative to the targets on the airfield. Many times, the exact locations of the aircraft are not known to the missile force, only the location of the potential parking locations due to delays in intelligence processing, missile transportation and preparation, missile launch, and the potentially long time of flight. Therefore, the aim points for the aircraft should be based on the parking locations and not the actual aircraft locations. The third step is to determine the impact location for each missile relative to its aim point using the missile's CEP and guidance probabilities. Finally, the last step is to determine the damage that results from the blast and fragmentation of the missiles for all nearby targets. Any targets within the missile–target lethal radius should be counted as damaged or destroyed. This lethal radius can be found from modeling and simulation software, such as the Joint Munitions Effectiveness Manual Weaponeering System (JWS), calculated based on vulnerability data[12] or observed from testing.

The lethal radii for the notional target–missile pairings used in the example test case are shown in Table 3. This work included three different types of aircraft and one type of POL station. TBM2 and CM1 are not included in the table because they carry runway-penetrating munitions, which have a very small lethal radius due to their subterranean detonation.

4. Air defense artillery simulation

ADA platforms defend against TBMs and CMs by detecting and destroying the incoming missiles with interceptor missiles, guns, or laser systems. Before the TBMs and CMs can impact their targets, they must first pass through the ADA defenses that can shoot down some or all of the incoming missiles. Missiles specified by the enemy fire plan are first tested against the ADA defenses using the ADA simulation, and those that are not shot down, the leakers, are passed to the damage effects programs.

In the example case for this paper, the Extended Air Defense Simulation (EADSIM) was used as the ADA simulation. The defending ADA forces were composed of two different notional systems: an Upper Tier (UT) system and a Lower Tier (LT) system. These systems were modifications of unclassified systems in the Demo300 demonstration scenario that is shipped with EADSIM. The UT system has a single interceptor type that can engage TBMs only. The LT system operates at altitudes below the UT system. It has three different interceptor types: Interceptors A, B, and C and can engage both TBMs and CMs. The notional specification of these intercept missiles is shown in Table 4. These representations were fabricated by the author and do not represent any known system.

The number of ADA systems and their assigned tactics must be set before optimizing the red fire plan. These settings should be based on the expectations of the attacking

Table 5. Notional base case air defense artillery system and tactics values.

System	Variable	Set value
Upper Tier	Number of launchers	6
	Maximum engagement altitude	60,000 meters
	Minimum engagement altitude	25,000 meters
	Shot doctrine	1
Lower Tier	Number of interceptor A launchers	2
	Number of interceptor B launchers	2
	Number of interceptor C launchers	4
	Maximum engaging altitude – TBMs	20,000 meters
	Minimum engagement altitude – TBMs	10,000 meters
	Maximum engagement range – CMs	5000 meters
	Minimum engagement range – CMs	2000 meters
	Shot doctrine – TBMs	2
	Shot doctrine – CMs	1
	Priority interceptor – TBMs	A
	Priority interceptor – CMs	C

TBM: theater ballistic missile; CM: cruise missile.

Table 6. Notional probability of kill (Pk) values for interceptor–target combinations.

Tier	Missile	Pk-TBM		Pk-CM	
		TBM1/2	TBM3	CM1	CM2
Lower	Int A	80	70	70	65
	Int B	60	50	85	80
	Int C	60	50	80	75
Upper	Upper	65	60	N/A	N/A

TBM: theater ballistic missile; CM: cruise missile.

missile forces. For this work, the notional simulation setup for the ADA systems and tactics are as shown in Table 5. The shot doctrine is the number of interceptors that are shot in rapid succession at a single inbound threat. The priority interceptor is the interceptor that the LT system will use against the given threat until exhausted. It will then use to the interceptor with the shortest time to intercept.

In most ADA simulations, the detection algorithm explicitly calculates the probability of detection based on the radar parameters and the target's radar cross-section. The rest of the kill chain, however, is calculated implicitly from a set of multi-dimensional probability of kill (Pk) tables, usually based on the geometry between the missile and the ADA system launcher. However, for simplicity in the example scenario, a single Pk was used for each interceptor–target pairing. These notional Pk values are summarized Table 6.

5. Optimization program

The optimization of the red fire plan is a complex task. Because a large section of the multi-dimensional design space is being explored, there is no guarantee that the design space will be convex. This means that gradient-based optimizers may not work and a stochastic optimizer is needed.[30] This work used the stochastic optimization software OptDef, which interfaces with an optimization engine called OptQuest, both of which are developed by OptTek Systems.[31] The OptQuest optimization engine uses a variety of metaheuristic techniques that include tabu search and scatter search as primary methodologies. The engine searches for global optima by iterating between diversification (exploring broadly) and intensification (finding local optima) phases.[31] OptQuest also slowly increases the precision it employs for each variable over the course of the optimization in order to explore finer levels of variable value changes and identify further unique solutions. Because there are competing objectives in the optimization, the final solution will be a set of non-dominated points called the Pareto frontier. This is the set of all points such that no other point is better in at least one optimization dimension than the non-dominated set. For problems with multiple objectives, OptQuest searches for the best solution in each objective direction and

attempts to push out the Pareto frontier by exploring gaps between frontier points.

In most optimization software applications, a separate script is required to interface between the optimization program and the simulation program or programs. This interface is commonly called a wrapper. For this work, the wrapper as well as the damage effects simulation were developed using MATLAB.[32] The wrapper function provides the mechanism to pass information between the ADA simulation, the damage effects simulation, and the optimization program. The process begins when the optimization program generates an initial run or reads the previous outputs and determines the next set of inputs in order to explore the non-dominated frontier. It then writes these inputs to a file and calls the wrapper. The general flow of the wrapper is as follows.

1. Replicate the input files of base ADA and damage effects simulations to a new folder location. In addition, create a folder for the ADA simulation and damage effects output files.
2. Read the inputs from the text file that the optimization program wrote and translate them into necessary inputs for the ADA simulation. For example, if a particular run specifies eight TBM1 missiles are to be launched, then the wrapper function must set eight TBM1 platforms to ''alive'' and the rest of the TBM1 platforms in the scenario to ''dead.''
3. Write the input values of the ADA simulation that have changed for the particular simulation run to the appropriate locations in the input files of the new folder created in step 1 while leaving the rest of the ADA simulation input file unchanged from the base simulation.
4. Call the ADA simulation program and specify the location of the new input files and the location to write the output files.
5. Read the ADA simulation output files and write the leaker information to the input files of the damage effects program at the location created in step 1.
6. Call the damage effects program by specifying the location of the input files and location to write the output values.
7. Extract the required output data from the damage effects program output files and write that to the output location specified by the optimization program.
8. If needed, delete the input and output files and folders that were created for the run in order to conserve disk space.
9. Report complete to the waiting optimization program.

The optimization can be easily completed in parallel, since each ADA simulation and subsequent damage effects simulation works with its own file set replicated from a base simulation. This work leveraged this parallel structure using multiple computer cores with the Department of Defense High Performance Computing Center (HPC).[33] To account for the stochastic nature of the ADA and damage effects program, the simulations for each optimization test point were repeated until the estimated mean of the output was within 10% of the true mean with 95% confidence.[34] For cases where the estimated mean was less than unity, the simulations were repeated until the estimated mean was within 0.1 of the true mean with 95% confidence. This mean was then reported to the optimization program as the result of the test case.

6. Example test case

Three fire plans of increasing size were developed to demonstrate the damage effects simulation paired with an ADA simulation. The first sample fire plan was a small attack made up of five TBMs and five CMs impacting the target simultaneously. These TBMs and CMs could be of any type (TBM1, TBM2, TBM3, or CM1 and CM2). The second fire plan was a medium sized attack made up of 15 TBMs and 15 CMs, of any type, impacting the target at the same time. Finally, the last fire plan was a large attack made up of 30 TBMs and 30 CMs, or any type, impacting the target simultaneously.

The three test cases were each optimized independently. The optimization criteria were to maximize the fraction (percentage) of aircraft destroyed, maximize the fraction of POL sites destroyed, and minimize the fraction of available MOSs. This multi-objective optimization created a frontier of non-dominated solutions for each fire plan. Each point on the frontier represents a combination of fire plan inputs that achieves a non-dominated solution in at least one of the optimization goals.

All fire plans included an identical setup for the input variables. The only difference between the fire plans was a constraint that ensured that the number of TBM and CM missiles matched the correct size for that fire plan. The fire plan input variables and ranges are shown in Table 7. Even though the largest fire plan allowed for 30 missiles, only a maximum of 20 missiles of any single type were allowed to account for the limitations in the number of launch platforms. For each type of missile, the fraction aimed at the runway was calculated as 100-(percent aimed at the airfield) and the fraction of airfield missiles aimed at the POL sites were calculated as 100-(percent to airfield missiles aimed at aircraft).

For the test case, a hypothetical airfield was created that has eight POL points spaced 50 meters apart in two

Table 7. Red fire plan optimization variables.

Input Variable	Type	Minimum	Maximum
Number of TBM1	Integer	0	20
Number of TBM2	Integer	0	20
Number of TBM3	Integer	0	20
Number of CM1	Integer	0	20
Number of CM2	Integer	0	20
TBM1 – percent aimed at airfield v. runway	Float	0	100
TBM2 – percent aimed at airfield v. runway	Float	0	100
TBM3 – percent aimed at airfield v. runway	Float	0	100
CM1 – percent aimed at airfield v. runway	Float	0	100
CM2 – percent aimed at airfield v. runway	Float	0	100
TBM release altitude for submunitions (meters)	Float	10,000	30,000
CM release altitude for submunitions (meters)	Float	50	500
Percent of airfield missiles aimed at aircraft v. POL points	Float	0	100
CM impact delay from TBM impact (seconds)	Float	−60	+60

TBM: theater ballistic missile; CM: cruise missile; POL: Petroleum, Oil, and Lubrication.

east–west rows of four points. The airfield also supported three different types of aircraft, AC1, AC2, and AC3. The parking locations for each aircraft were 50 meters apart. Aircraft type 1 (AC1) was arranged in two rows of 15 parking spots angled parallel to the runway. Aircraft type 2 (AC2) was set further south of the parking for aircraft type 1 in a single row of 15 spots. The parking for aircraft type 3 (AC3) was set just north-west of the other parking locations in two east–west rows of six parking spots each. This notional airfield is shown in Figure 8.

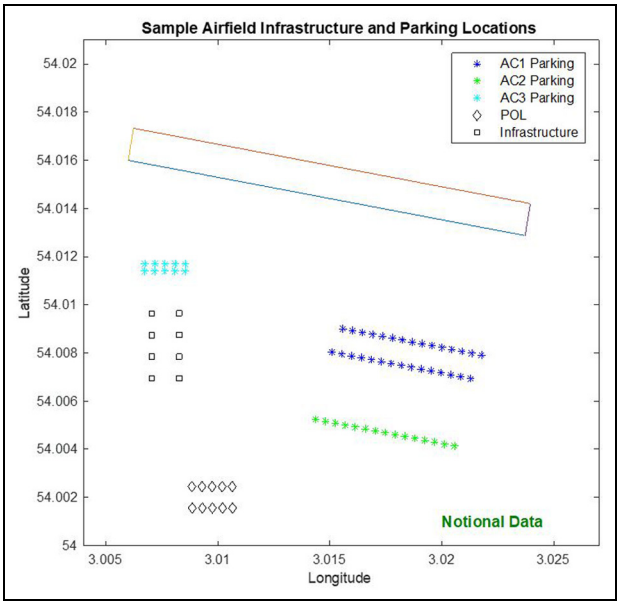


Figure 8. Notional airfield with Petroleum, Oil, and Lubrication (POL) and aircraft parking locations.

7. Optimization results

The results of the optimization for the small, medium, and large fire plans are shown in Figures 9 and 10(a). These images show the frontier of non-dominated solutions. The optimal frontier is further illustrated in Figure 10(b), which shows all the tested points for the large attack, to include the dominated test points, along with the optimal frontier. In Figures 9 and 10, the two horizontal axes are the fraction of aircraft destroyed and fraction of POL sites destroyed. The vertical axis is the fraction of MOSs available on the runway for use. The missile force is trying to maximize the fraction of aircraft and POL sites destroyed and minimize the fraction of MOSs available. Thus, the optimal point is the lower-right point of each graph.

As the size of the fire plans increased from small to large, the non-dominated front's curvature also increased. The frontier also moved toward the lower-right of the graph, indicating that the increased number of leakers from the larger sized fire plans caused greater damage to the airfield. The large number of points on the frontier of all three sized fire plans shows how there is a tradeoff for damage to the various parts of the airfield. Certain fire plans caused a greater number of specific types of leakers, which, in turn, caused greater damage to specific parts of the airfield. This is illustrated in Table 8.

Table 8 shows two points on the optimal frontier for the medium-sized fire plan. Each of these points used a fire plan composed of 15 TBMs and 15 CMs; however, the outputs from each were different based on the type of missiles, targets, and attack parameters. Sample point 1 in Table 8 is composed of a mix of TBM2s and TBM3s and CM1s and CM2s. All but one of the TBMs was aimed at the runway and the TBMs with submunitions released those submunitions at the maximum allowable altitude.

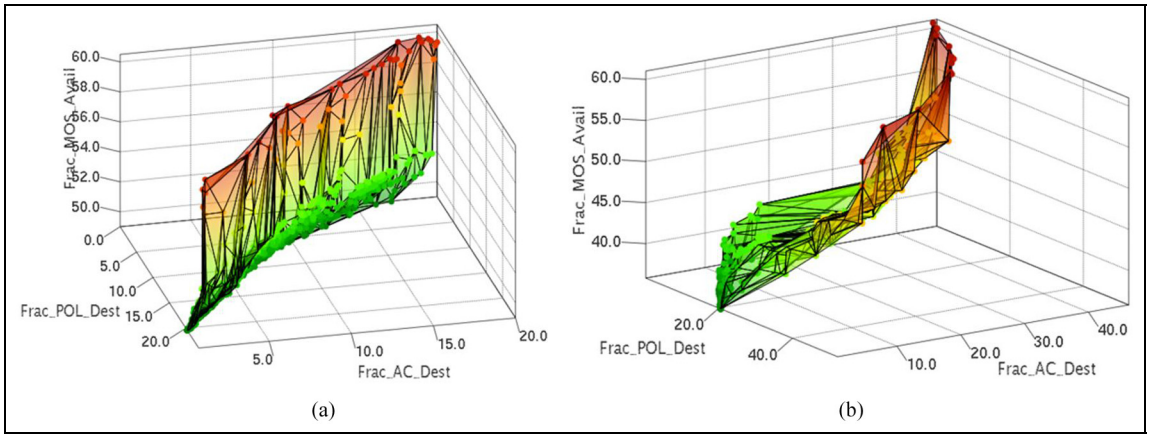


Figure 9. Optimal frontier for small-sized fire plan (a) and medium-sized fire plan (b).

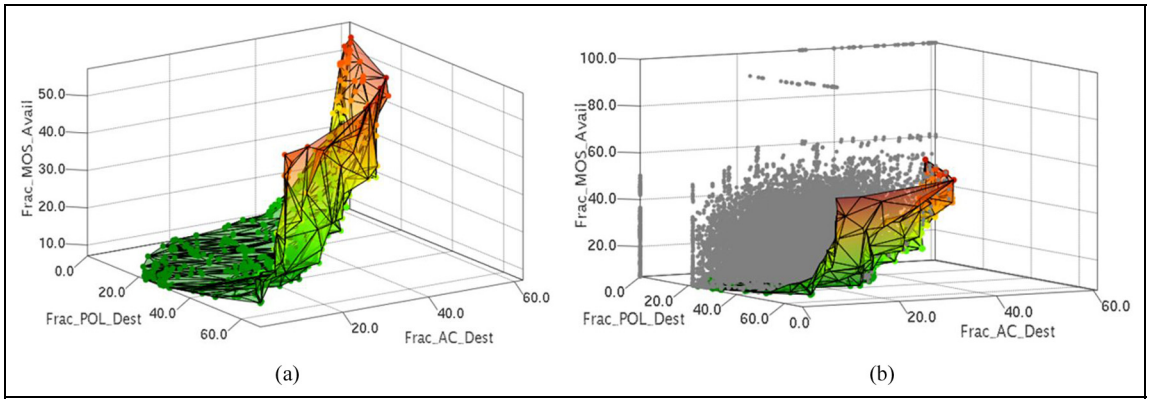


Figure 10. Optimal frontier for large-sized fire plan (a) and large-sized fire plan with both dominated and frontier points (b).

This reduced their vulnerability to the ADA system. CM1 released its submunitions at a higher altitude that increased its spread around the runway, but this made it more vulnerable to the ADA systems since it was flying at a higher altitude. To counter this disadvantage, the CMs were timed to arrive 3 seconds before the TBMs. This meant that the ADA systems were likely engaging the TBMs when the CMs first crossed the radar horizon, reducing the ADA systems engagement window against the CMs. This resulted in a smaller portion of usable runway following the attack compared with sample point number 2 in Table 8.

Contrast this with sample point number 2 in Table 8, which is a uniform attack consisting of only a single type of TBM and CM: TBM3 and CM1 missiles. All of the CM1 missiles were aimed at the runway and released their submunitions at a very low altitude arriving 7 seconds after the TBMs. Most of the TBM3 missiles were aimed at the airfield and were concentrated on the parked aircraft. This resulted in a larger portion of usable runway still available, but with a greater number of aircraft and POL points destroyed in the attack, compared to sample point number 1 in Table 8.

8. Conclusions and future work

This work demonstrated a methodology to quantify the damage from missiles that leak through the ADA defenses and impact an airfield. It also demonstrated a way to calculate the frontier of optimal fire plans that maximize damage to different parts of an airfield, such as the POL sites, parked aircraft, or runway. Depending on the analysis, the analyst may want to include all of these optimal fire plans in subsequent analysis or use a multi-attribute decision making technique (MADM) to choose a single or smaller subset of the fire plans in the subsequent analysis. In either case, a cost model could be included in the analysis so that the fire plan is optimized for the damage caused to the airfield at the lowest cost.

Table 8. Sample points on optimal frontier for medium-sized fire plan.

	Variable		Sample frontier point	
	Input/output		1	2
Input	Number of missiles in fire plan	TBM1	0	0
		TBM2	13	0
		TBM3	2	15
		CM1	9	15
		CM2	6	0
	Percent of missiles aimed at airfield v. runway	TBM1	0	0
		TBM2	0	0
		TBM3	50	94
		CM1	11	100
		CM2	0	0
	TBM release altitude for submunitions (meters)		30,000	10,000
	CM release altitude for submunitions (meters)		491	105
	Percent of airfield missiles aimed at aircraft v. POL points		66%	85
	CM impact delay from TBM impact (seconds)		−3	7
Output	Fraction of MOSs remaining after attack		40.7%	60.7%
	Fraction of POL points destroyed in attack		5.3%	11.6%
	Fraction of aircraft destroyed in attack		10.9%	40.9%
	Number of missiles that impact the ground (leakers)	TBM1	0	0
		TBM2	3.9	0
		TBM3	0.3	5.5
		CM1	3.0	1.9
		CM2	2.9	0

TBM: theater ballistic missile; CM: cruise missile; POL: Petroleum, Oil, and Lubrication; MOS: minimum operating strip.

Optimization of the missile fire plan should be performed before any optimization or analysis of the ADA systems or tactics. However, as the ADA systems and tactics change, the missile fire plan must be optimized once again. This is because the method shown in this work optimizes the missile fire plan parameters based on a predetermined number of ADA systems and tactics. By coupling this analysis to an ADA systems and tactics optimization, the analyst could engage in a two-sided optimization with the missile fire plan optimization occurring first followed by the optimization of the ADA system and tactics. This would continue until convergence.

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References

1. Gormley DM. *Missile contagion: cruise missile proliferation and the threat to international security*. Annapolis, MD: Naval Institute Press, 2008.
2. Feickert A. *Cruise missile proliferation*. Washington DC: Congressional Research Service, The Library of Congress, 2005.
3. Pollack J. Ballistic trajectory: the evolution of North Korea's ballistic missile market. *Nonprolif Rev* 2011; 18: 411–426.
4. Ahuja RK, Kuma A, Jha KC, et al. Exact and heuristic algorithms for the weapon-target assignment problem. *Oper Res* 2007; 55: 1136–1146.
5. Wu L, Xing C, Lu F, et al. An anytime algorithm applied to dynamic weapon-target allocation problem with decreasing

weapons and targets, In: *2008 IEEE congress on evolutionary computation*, Hong Kong, 1 June 2008, pp.3755–3759. Piscataway, NJ: IEEE.
6. Ender T, Leurck RF, Weaver B, et al. Systems-of-systems analysis of ballistic missile defense architecture effectiveness through surrogate modeling and simulation. *IEEE Syst J* 2010; 4: 156–166.
7. Holland OT and Wallace SE. Using agents to model the kill chain of the ballistic missile defense system. *Nav Eng J* 2011; 123: 141–151.
8. Parnell GS, Metzger RE, Merrick J, et al. Multiobjective decision analysis of theater missile defense architectures. *Syst Eng* 2001; 4: 24–34.
9. Wade BM and Chang P. New measures of effectiveness for the air and missile defense simulation community. *Phalanx* 2015; 48: 49–53.
10. Gormley DM. The neglected dimension: controlling cruise missile proliferation. *The Nonprolif Rev* 2002; 9, pp. 21–29.
11. Brown G, Carlyle M, Diehl D, et al. A two-sided optimization for theater ballistic missile defense. *Oper Res* 2005; 53: 745–763.
12. Driels MR. *Weaponeering: conventional weapon system effectiveness*. Reston, VA: American Institute of Aeronautics and Astronautics, 2004.
13. Chabai AJ. On scaling dimensions of craters produced by buried explosives. *J Geophys Res* 1965; 70: 5075–5098.
14. Kiger SA and Balsara JP. *A review of the 1983 revision of TM 5–855–1 ''Fundamentals of Protective Design'' (nonnuclear)*. Vicksburg, MS: USAE Waterways Experiment Station, 1983.
15. *Ballistic and cruise missile threat*. Wright-Patterson AFB, OH: National Air and Space Intelligene Center, 2013.
16. Forden G. GUI missile flyout: a general program for simulating ballistic missiles. *Sci Global Secur* 2007; 15: 133–146.
17. Anderson JD. *Aircraft performance and design*. Boston, MA: McGraw-Hill, 1999.
18. Carlucci DE and Jacobson SS. *Ballistics: theory and design of guns and ammunition*. 2nd ed. Boca Raton, FL: Taylor & Francis Group, 2014.
19. McCoy RL. *Modern exterior ballistics: the launch and flight dynamics of symmetric projectiles*. 2nd ed. Atglen, PA: Schiffer Military History, 2012.
20. Yu R, Spiesz P and Brouwers HJH. Numerical simulation of ultra-high performance fibre reinforced concrete (UHPFRC) under high velocity impact of deformable projectiles. In: *15th international symposium on interaction of the effects of munitions with structures*, Potsdam, Germany, 16–20 September 2013, pp. 1–10. Amsterdam: Elsevier.
21. Ben-Dor G, Dubinsky A and Elperin T. Ballistic impact: recent advances in analytical modeling of plate penetration dynamics - a review. *Appl Mech Rev* 2005; 58: 355–370.
22. Wade B. A Comparison of three empirical methods for concrete penetration depth by a high speed ogive shaped projectile. In: *proceedings of the 6th annual world conference of the society for industrial and systems engineering*, Herndon, VA, 19–20 October 2017. The Society for Industrial and Systems Engineering.
23. Young CW. *Penetration equations*. Albuquerque, NM: Sandia National Laboratories, 1997.
24. DiGiulian T. Calculating CRH, http://www.navweaps.com/index_tech/tech-094.htm (2004, accessed 29 April 2015).
25. Kumaran US, Lim CT, Ong KCG, et al. Effect of different nose-shaped projectiles on the high speed perforation of concrete. In: Furnish MD, Chhabildas LC and Hixson RS (eds) *Shock compression of condensed matter*. Melville, NY: American Institute of Physics, 1999, pp.1233–1236.
26. Bening RG and Kurtz MK. *The formation of a crater as observed in a series of laboratory-scale cratering experiments*. Livermore, CA: US Army Engineer Nuclear Cratering Group, 1967.
27. Bull JW and Woodford CH. Camouflets and their effect on runway support. *Comput Struct* 1998; 69: 695–706.
28. Maienschein JL. Estimating equivalency of explosives through a thermochemical approach, preprint. In: *12th international conference symposium*, San Diego, CA, 11–16 August 2002.
29. *Airfield damage repair operations*. Washinton DC: Department of the Air Force, 2008.
30. Vanderplaats GN. *Multidiscipline design optimization*. Colorado Springs, CO: Vanderplaats Research and Development, Inc., 2007.
31. OptTek Systems, http://www.opttek.com/optdef (accessed 3 March 2017).
32. MATLAB, https://www.mathworks.com/products/matlab.html (accessed 3 March 2017).
33. The Depatment of Defense High Performance Computing (HPC) Environment, https://centers.hpc.mil (accessed 31 January 2017).
34. Law AM. *Simulation modeling and analysis*. 5th ed. New York: McGraw Hill Education, 2015.

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